NAAN MUDHALVAN PROJECT

COURIER MANAGEMENT SYSTEM(CMS)

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1.Introduction:

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies. The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

1.1Purpose of CMS:

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability.

2.REQUIREMENT ANALYSIS:

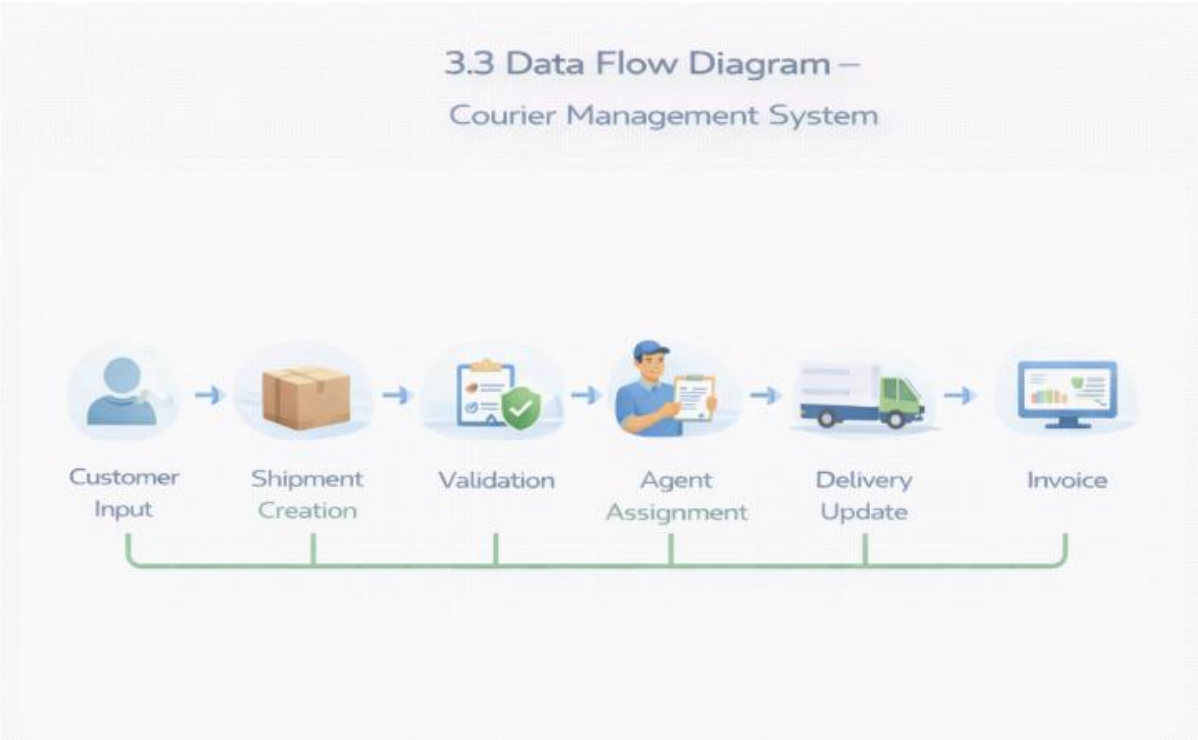
Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation

Non-Functional Requirements

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.DATA FLOW DIAGRAM:



4.PROJECT DESIGN PHASE :

Problem–Solution Fit CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.

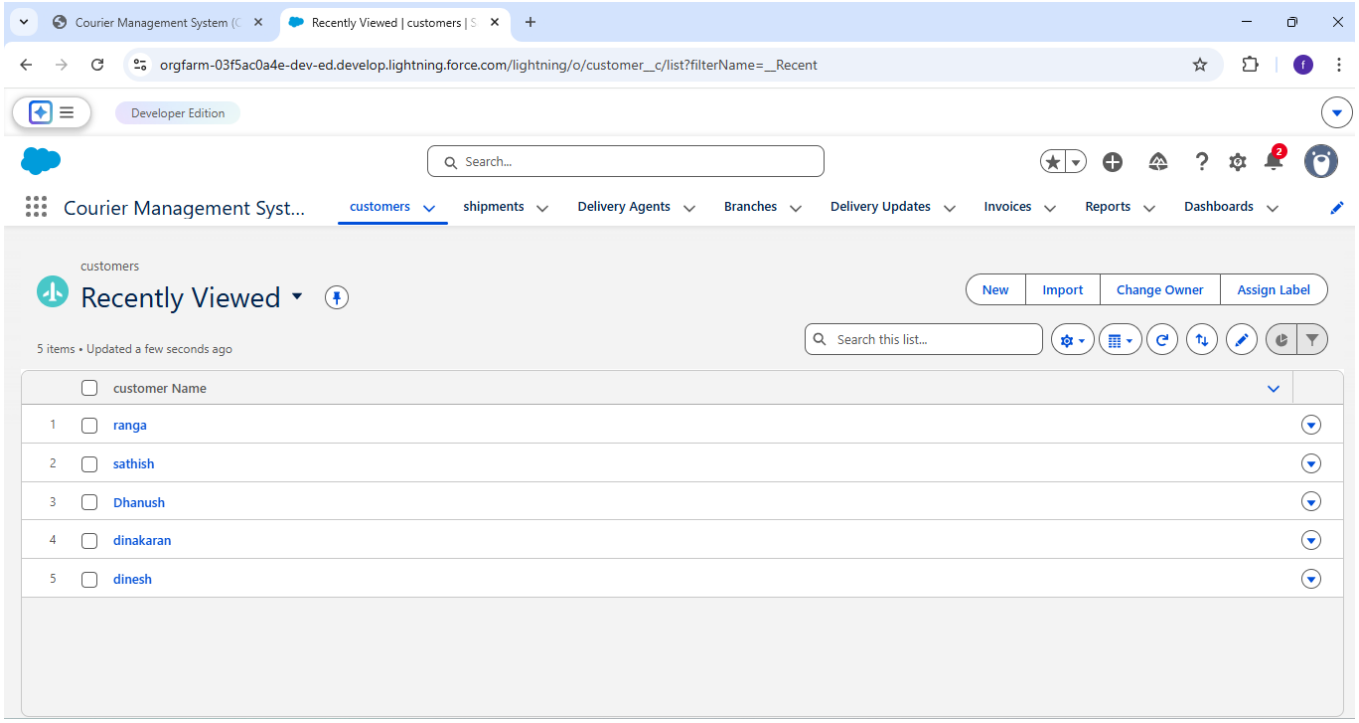
4.1.Solution Architecture Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

5.PROJECT DEPLOYMENT PHASE:

5.1Courier management system overview:

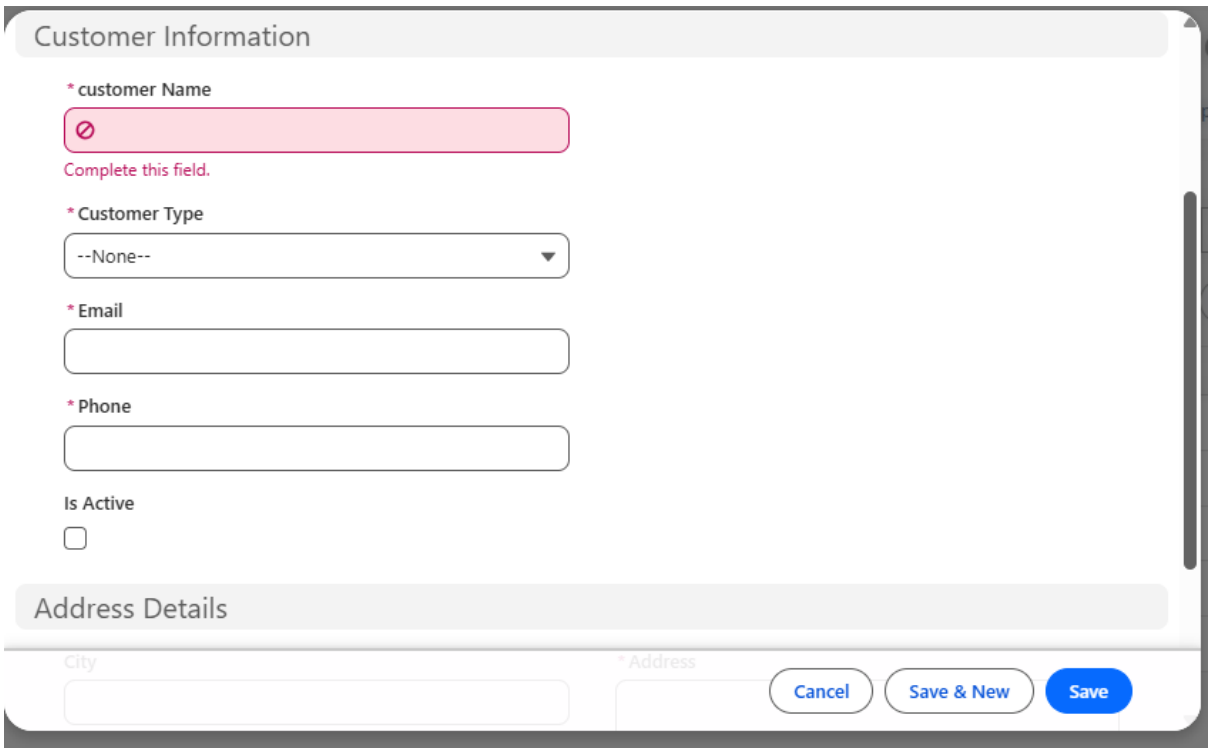
The Courier Management System is a software application designed to automate, track, and manage end-to-end logistics operations, including booking, courier routing, real-time parcel tracking, and, staff/branch management. It replaces manual, error-prone processes with a centralized digital system, improving delivery efficiency, reducing costs, and enhancing customer service



5.2.Customers tab :

The Customer Tab within the Courier Management Service project serves as a centralized CRM module designed to efficiently manage, track, and update client-specific information. It enables staff to add, view, and modify customer profiles, including contact details, addresses, and account statuses, which reduces manual paperwork and streamlines booking operations. The module provides instant access to a client's comprehensive history of bookings, allowing for quick retrieval of past shipments and associated billing details. Furthermore, this tab enhances customer experience by facilitating real-time tracking, enabling service representatives to promptly handle inquiries, and storing

preferences for repeat business, thereby improving overall operational efficiency and strengthening customer satisfaction.

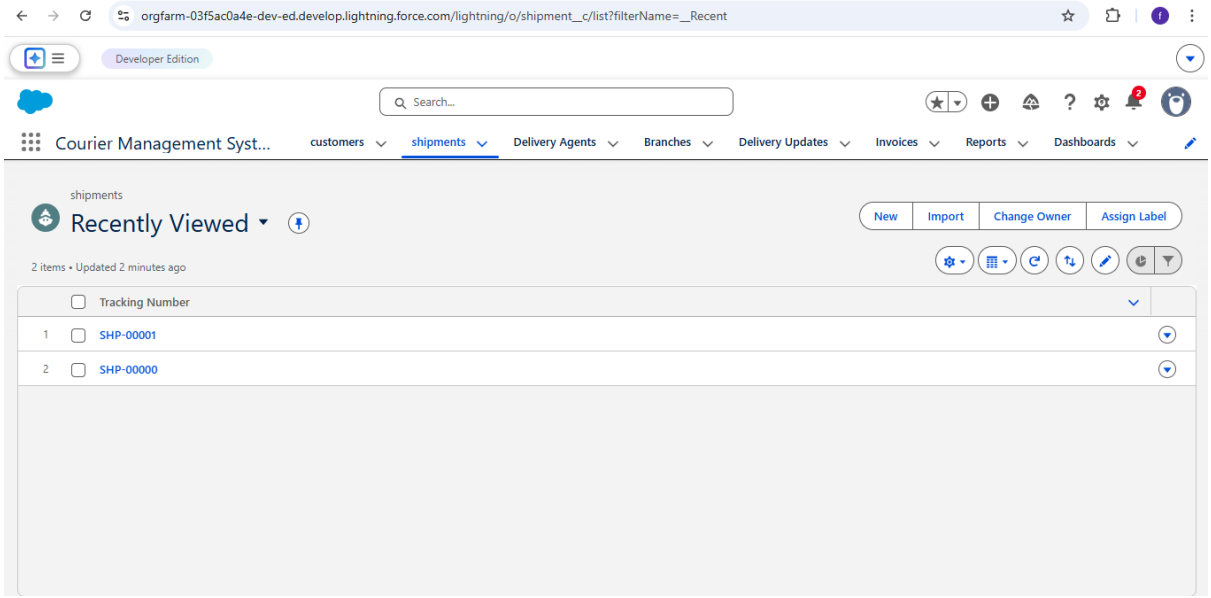


In this field we can add customers details like

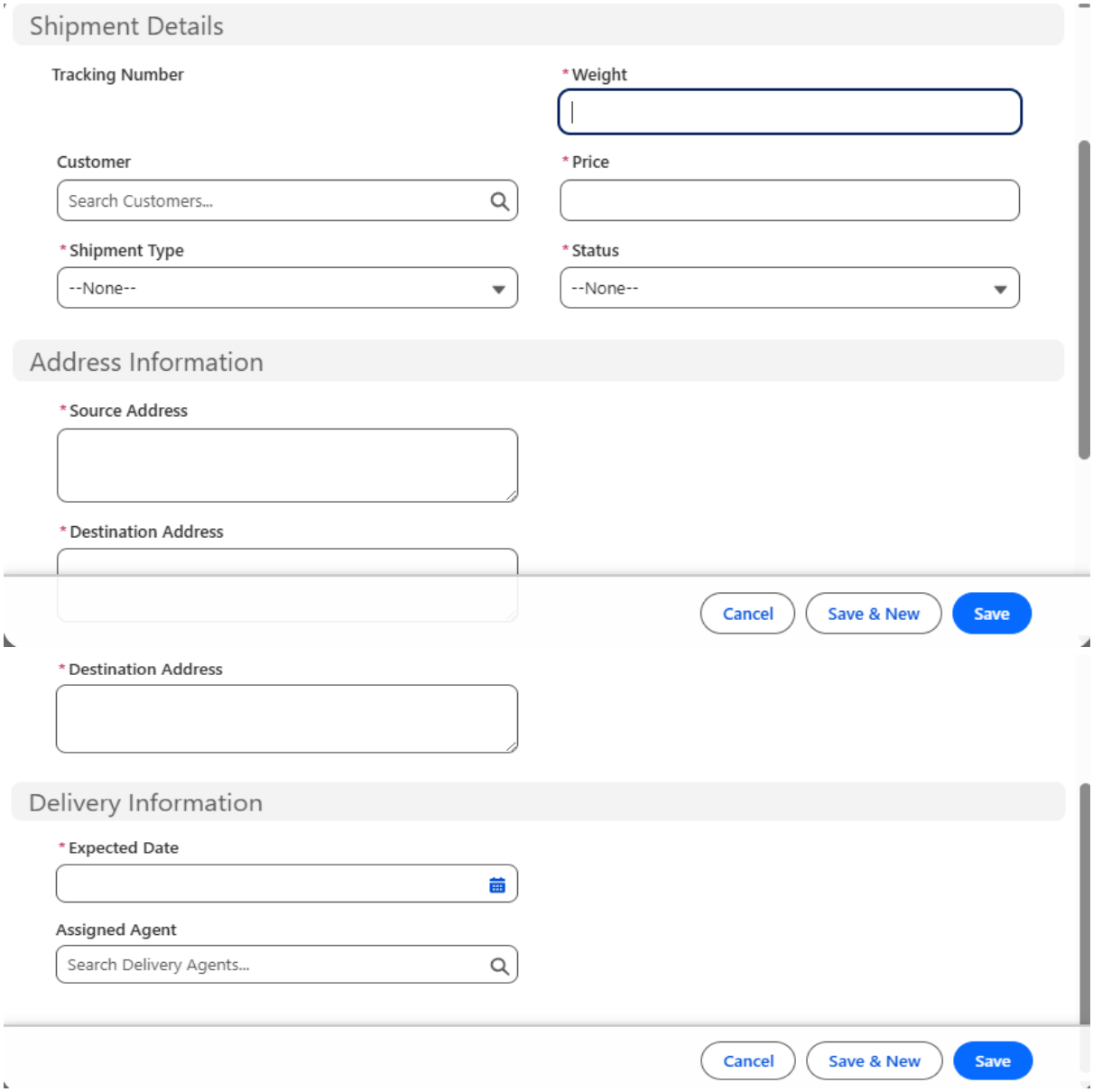
- Customer name
- Customer type
- Email id
- Phone
- City
- Address
- State
- Pincode

5.3.Shipment tab:

The Shipment Tab serves as the central command center for managing the entire lifecycle of courier consignments, enabling users to create, monitor, and update package details from a single, intuitive interface. Designed to streamline operations, this tab allows for efficient, real-time tracking of packages from origin to destination, providing immediate updates on transit statuses such as "Shipped," "In-Transit," "Out for Delivery," or "Delivered". Within this tab, users can generate unique Tracking IDs (AWB), print shipping labels, calculate courier rates, and upload Proof of Delivery (POD) documents, including digital signatures, to ensure accountability and reduce paperwork. Additionally, the tab features advanced filtering options for searching specific shipments by tracking number, status, or date, and facilitates the management of both individual and bulk shipments to boost overall operational efficiency and improve customercommunication.



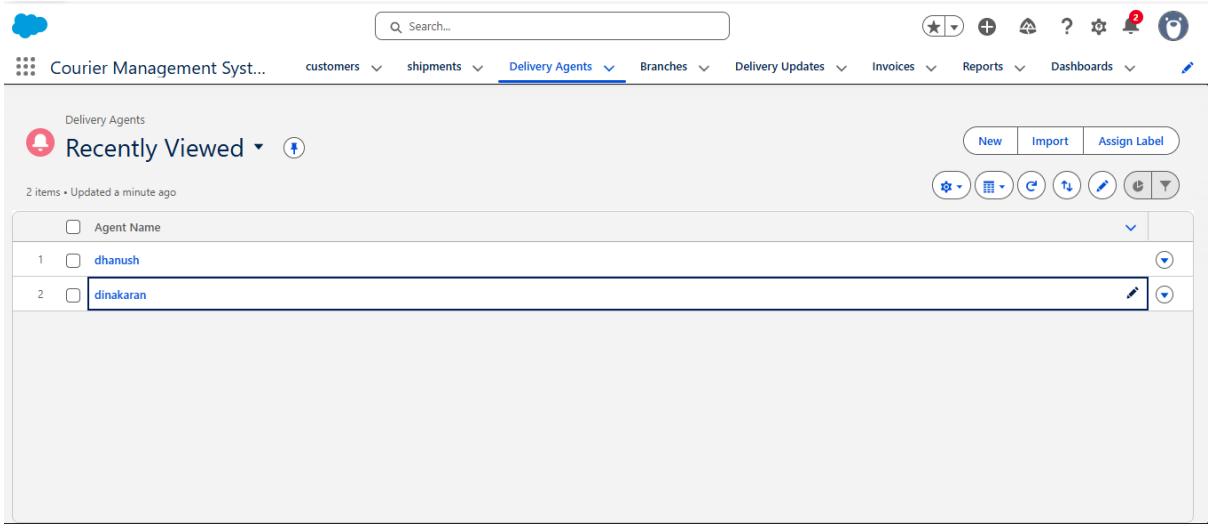
Shipment detials



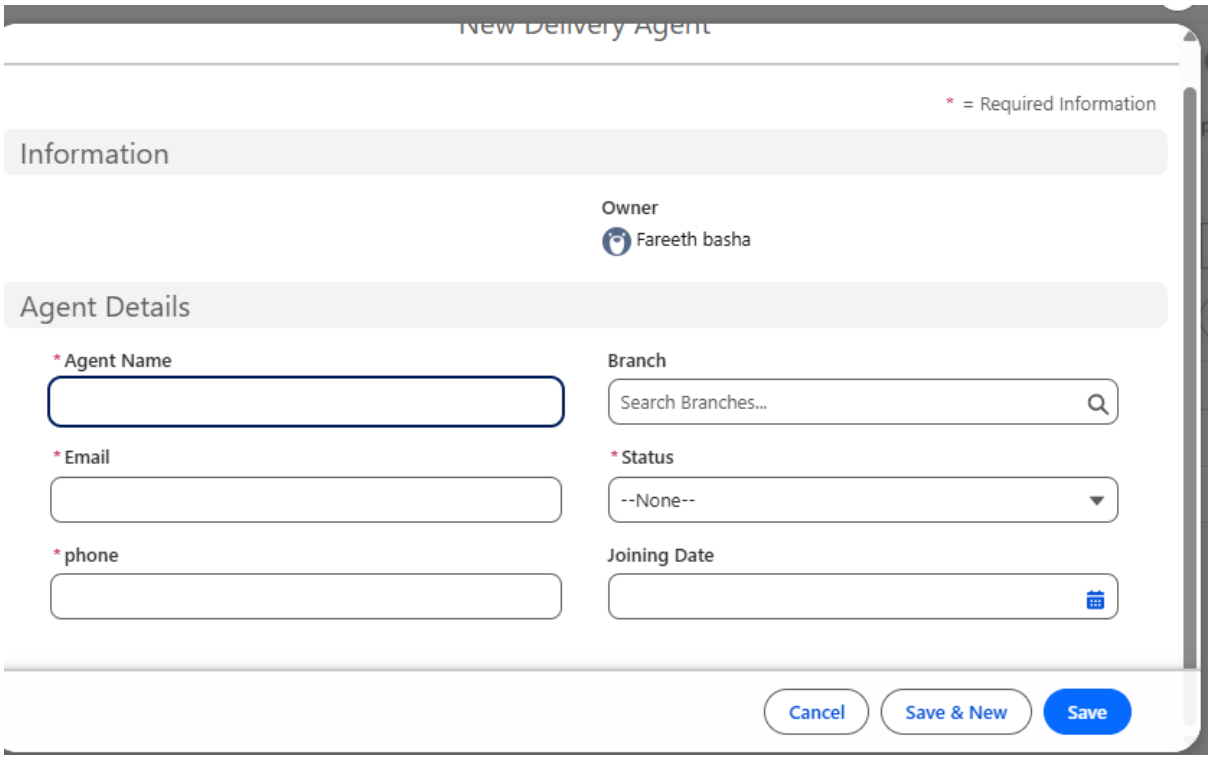
5.4.Delivery agents tab:

The **Delivery Agents** tab serves as the central hub for managing the last-mile workforce, allowing administrators to onboard, monitor, and optimize the performance of delivery personnel in real time. This module provides a comprehensive overview of agent availability, active assignments, and location data, facilitating efficient dispatching through smart assignment tools based on proximity and capacity. It enables managers to view

individual agent profiles, including contact details, vehicle information, and performance metrics such as on-time delivery rates, while also allowing for the management of attendance and salary calculations. Through this tab, administrators can track active deliveries on a map, view Proof of Delivery (POD) data (signatures/photos) for completed tasks, and instantly assign new pickup or delivery requests, ensuring high accountability and improved operational efficiency.

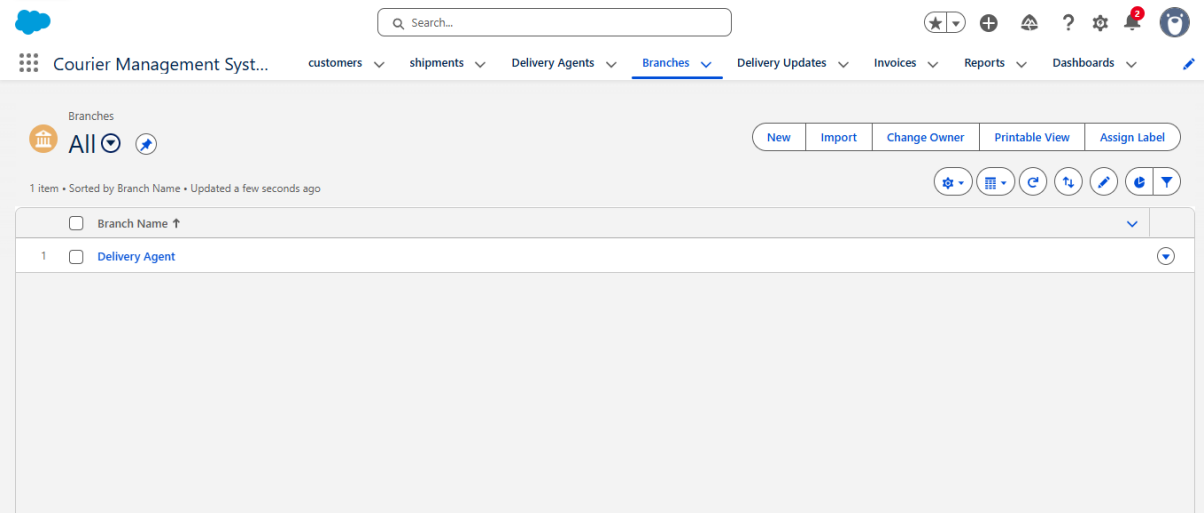


Adding new delivery agents:

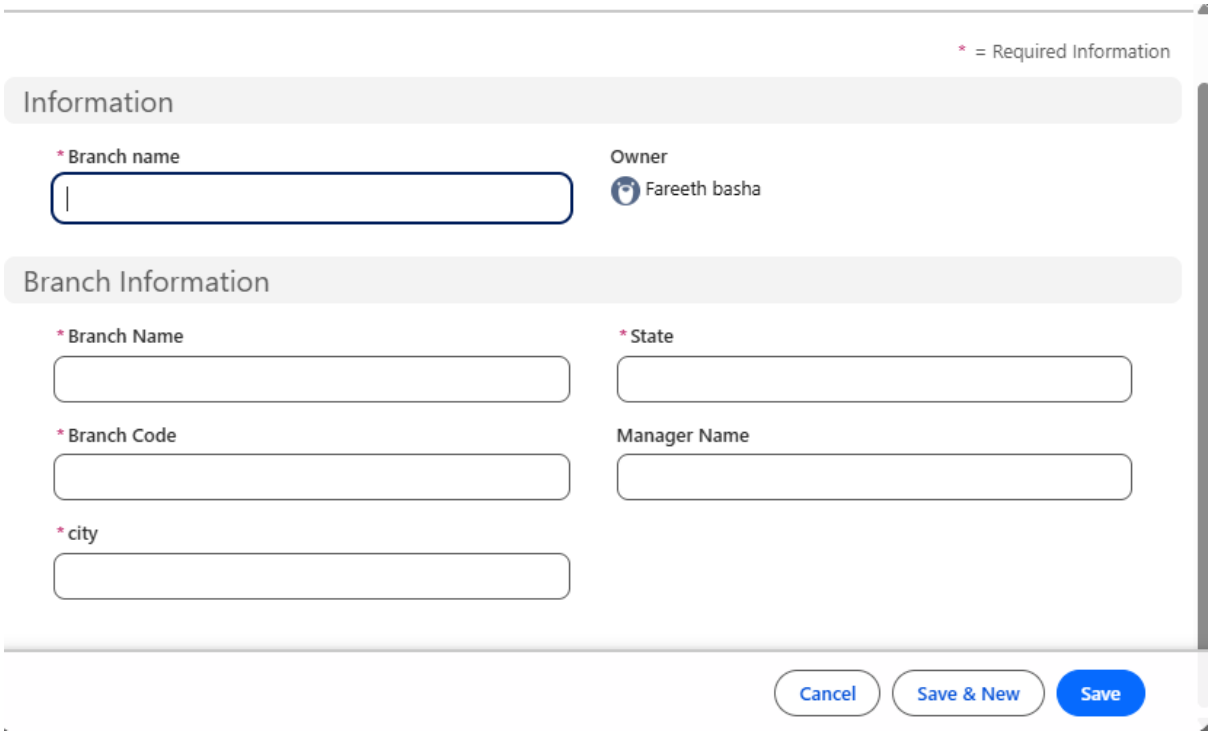


5.5Branches Tab:

The **Branches Tab** serves as the central administrative hub for managing the physical infrastructure of the courier service. It enables administrators to create, update, and monitor various branch locations, including details like branch name, location, address, contact information, and branch code. This module is crucial for organizing the logistical network, assigning staff to specific branches, and managing local inventory or incoming/outgoing shipments. Through this tab, users can view a comprehensive list of all operational branches, facilitating efficient, centralized control over the entire network from a single interface, ensuring seamless coordination and reducing errors in logistical operations.



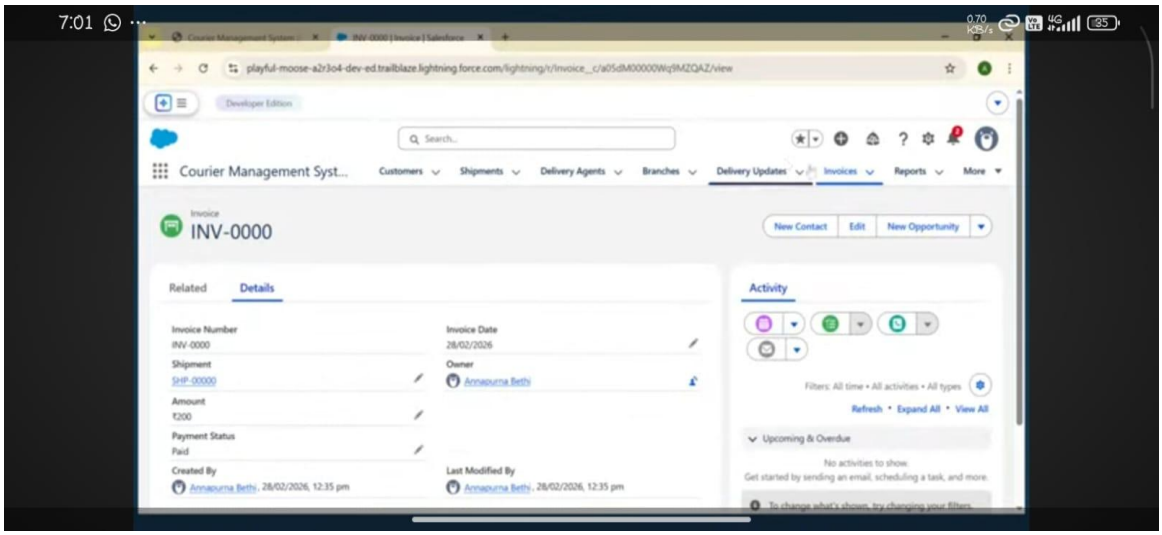
Creating new branch:



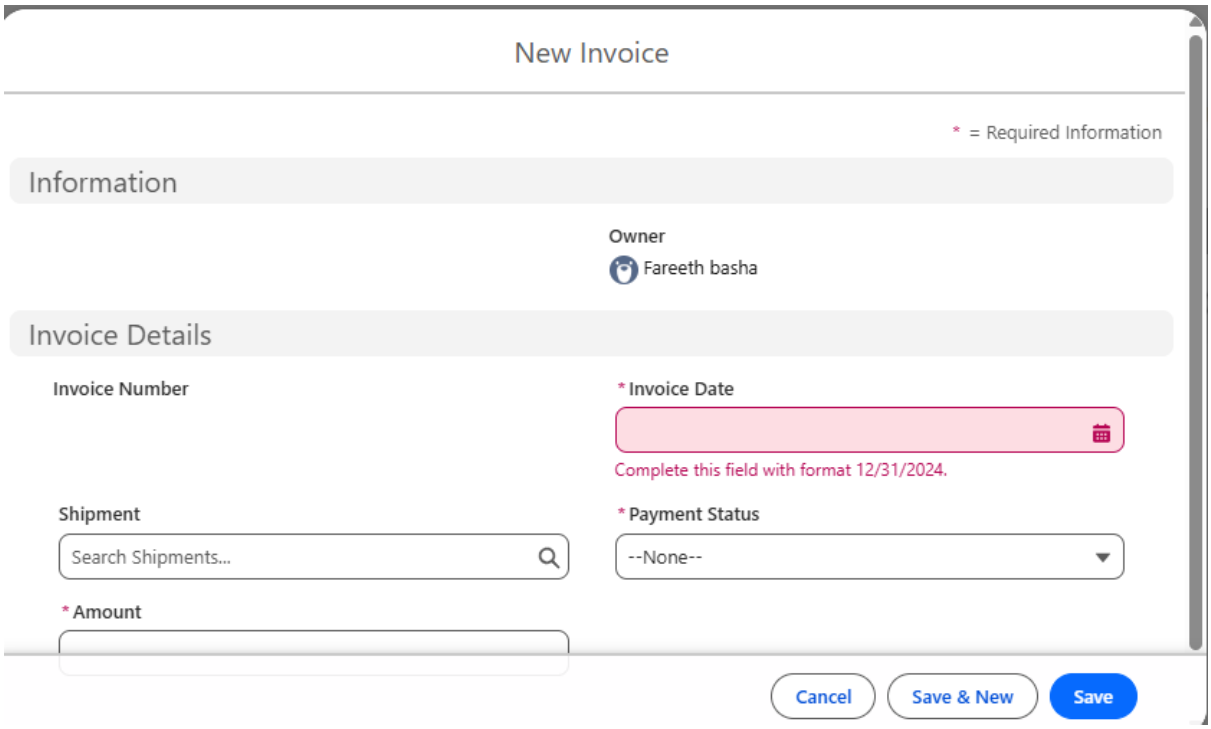
5.6Invoice tab:

The Invoice Tab in the Courier Management Service project serves as the central hub for financial transactions, enabling the automated generation, tracking, and management of customer billing. It allows administrators to quickly create professional invoices based on shipment data—including weight, destination, and service type—ensuring accurate calculations for freight, handling, and applicable GST. Key functionalities include managing invoice status (paid, unpaid, or overdue), generating weekly or monthly reports, and offering customizable, branded invoice templates to enhance professionalism. Furthermore, the tab supports secure, real-time payment tracking, facilitating faster,

streamlined cash flow and minimizing manual data entry errors.



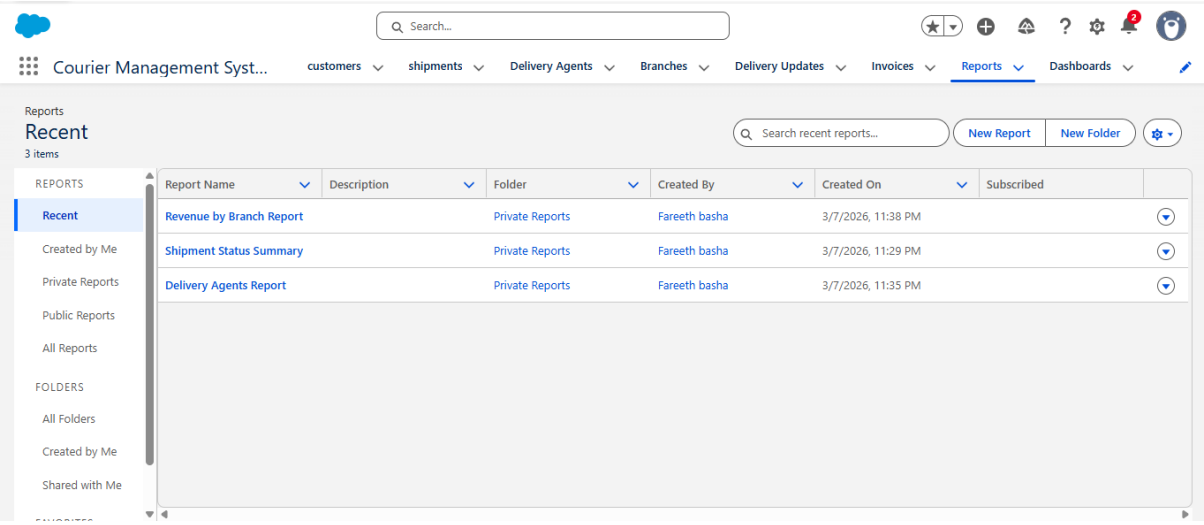
Creating new invoice:



5.7Report tab:

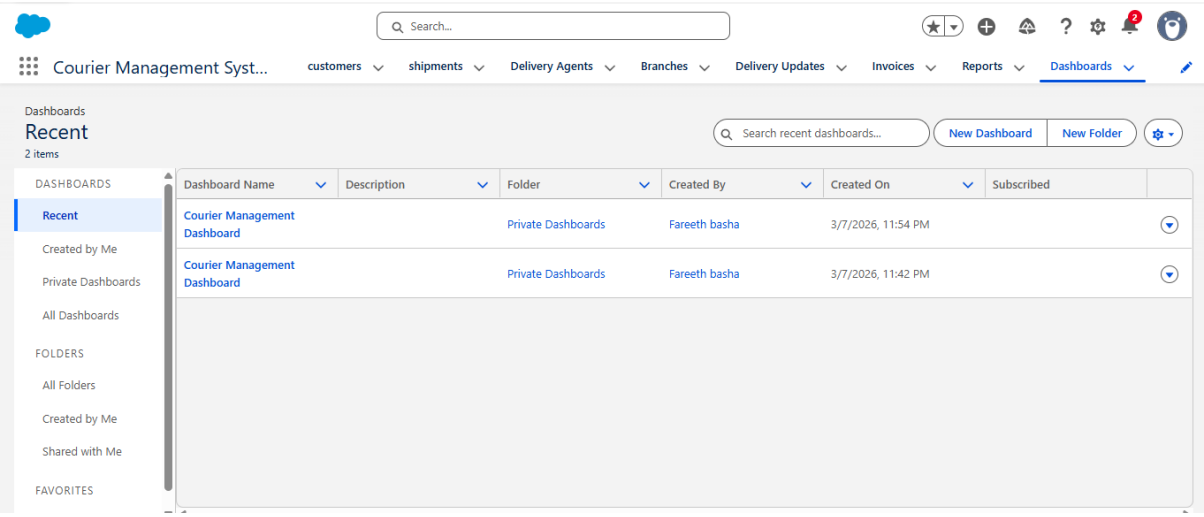
The Reporting module serves as the analytical hub of the Courier Management System, empowering administrators and managers to generate comprehensive reports on daily, weekly, or monthly operations with just a few clicks. This tab provides detailed insights into various logistical aspects, including booking records,

incoming/outgoing shipments, branch-wise performance, and courier status updates (delivered, pending, or returned). By facilitating the generation of automated invoices, delivery reports, and financial summaries, it eliminates the need for manual record-keeping, thereby minimizing human error and saving significant operational time. Furthermore, the system allows for the analysis of courier data to identify bottlenecks, track employee performance, and monitor revenue, providing actionable data to optimize overall operational efficiency.



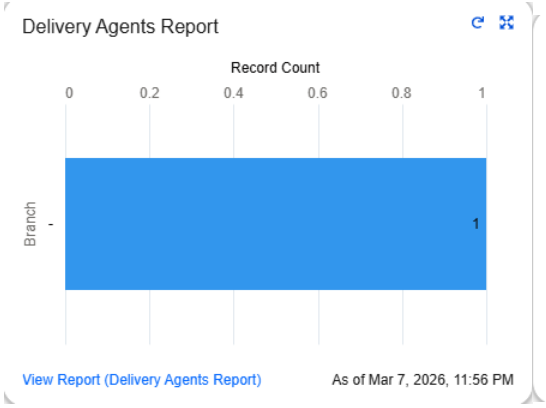
5.8.Dashboard tab:

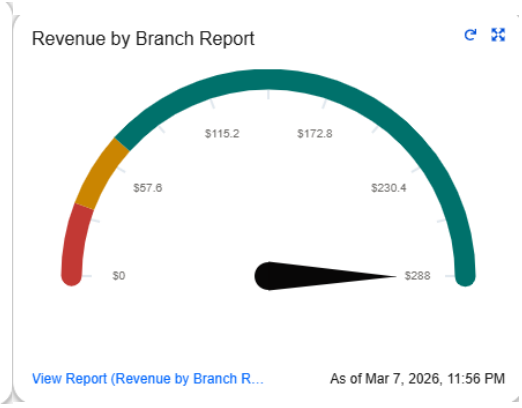
The Dashboard tab serves as the central command center for all courier operations, providing a comprehensive, real-time bird's-eye view of your entire delivery network. It aggregates crucial data, including total pending and delivered orders, active driver tracking, and instant alerts on delayed shipments to ensure proactive management. Through an intuitive, data-driven interface, administrators can monitor KPIs like on-time delivery rates, revenue generated, and regional performance trends, empowering them to make faster, informed decisions that boost operational efficiency and enhance customer satisfaction



Final Layout :

6.Final result:





7.ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

8.FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

9.Conclusion:

The implementation of the Salesforce-based Courier Management Service project successfully modernizes logistics operations by transitioning from manual, disparate processes to a centralized, automated, and scalable cloud platform. By leveraging Salesforce's robust CRM capabilities, the system provides real-time tracking, enhanced customer service through automated notifications, and improved operational efficiency, significantly reducing human error and delivery times.

Thank you!